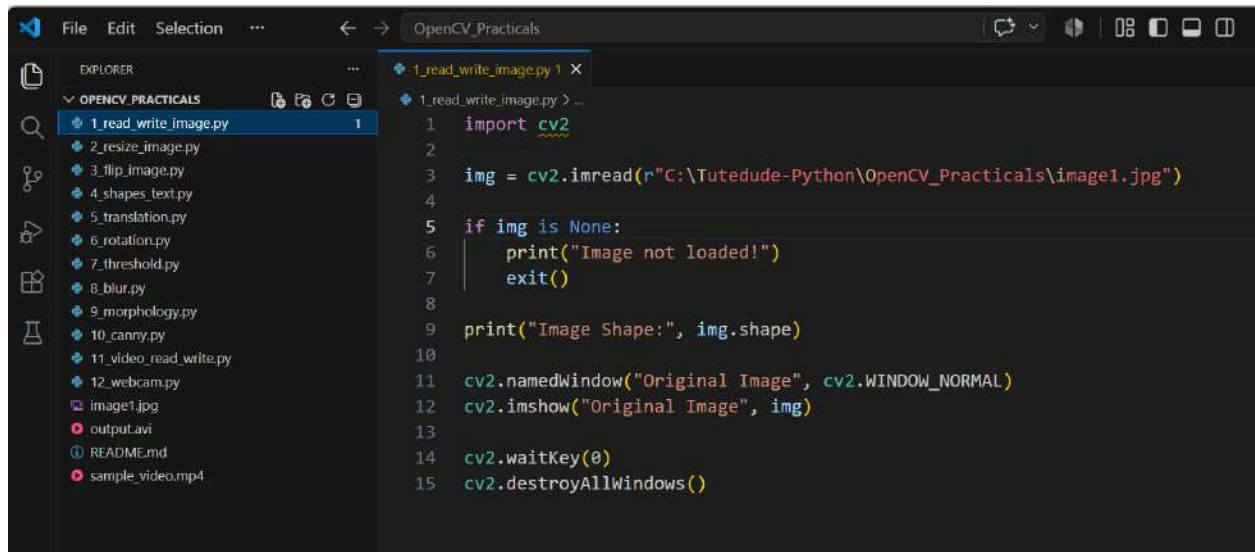


# Assignment- OpenCV Practicals



The screenshot shows a code editor window titled "OpenCV\_Practicals". The left sidebar displays a file explorer with a list of Python files: 1.read\_write\_image.py, 2.resize\_image.py, 3.flip\_image.py, 4.shapes\_text.py, 5.translation.py, 6.rotation.py, 7.threshold.py, 8.blur.py, 9.morphology.py, 10.canny.py, 11.video\_read\_write.py, 12.webcam.py, image1.jpg, output.avi, README.md, and sample\_video.mp4. The main editor area shows the code for 1.read\_write\_image.py, which reads an image from a specified path, checks if it was loaded, prints its shape, and displays it in a window titled "Original Image".

```
1 import cv2
2
3 img = cv2.imread(r"C:\Tutedude-Python\OpenCV_Practicals\image1.jpg")
4
5 if img is None:
6     print("Image not loaded!")
7     exit()
8
9 print("Image Shape:", img.shape)
10
11 cv2.namedWindow("Original Image", cv2.WINDOW_NORMAL)
12 cv2.imshow("Original Image", img)
13
14 cv2.waitKey(0)
15 cv2.destroyAllWindows()
```

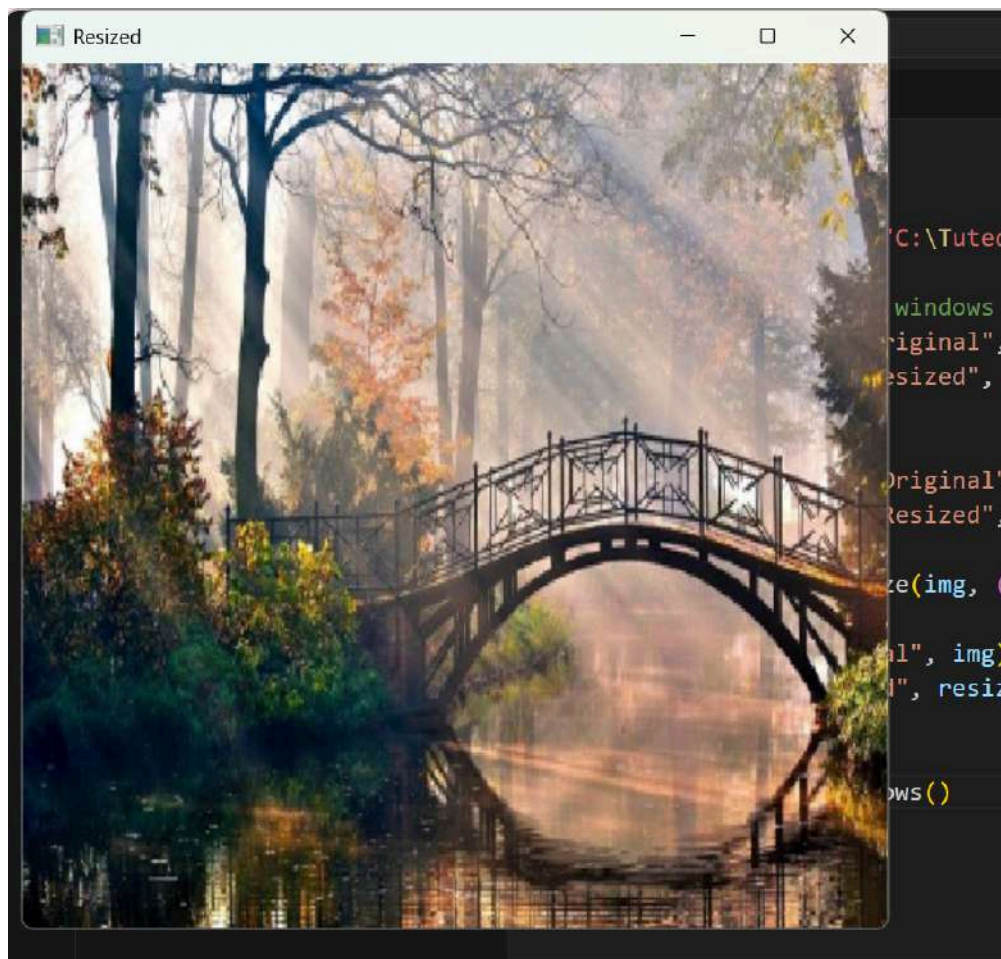
Output:



```
File Edit Selection ... OpenCV_Practicals
EXPLORER
OPENCV_PRACTICALS
1_read_write_image.py
2_resize_image.py
3_flip_image.py
4_shapes_text.py
5_translation.py
6_rotation.py
7_threshold.py
8_blur.py
9_morphology.py
10_canny.py
11_video_read_write.py
12_webcam.py
image1.jpg
output.avi
README.md
sample_video.mp4

2_resize_image.py
1 import cv2
2
3 img = cv2.imread(r"C:\Tutedude-Python\OpenCV_Practicals\image1.jpg")
4
5 # Create resizable windows
6 cv2.namedWindow("Original", cv2.WINDOW_NORMAL)
7 cv2.namedWindow("Resized", cv2.WINDOW_NORMAL)
8
9 # Set window sizes
10 cv2.resizeWindow("Original", 500, 500)
11 cv2.resizeWindow("Resized", 500, 500)
12
13 resized = cv2.resize(img, (400, 300))
14
15 cv2.imshow("Original", img)
16 cv2.imshow("Resized", resized)
17
18 cv2.waitKey(0)
19 cv2.destroyAllWindows()
```

Output:

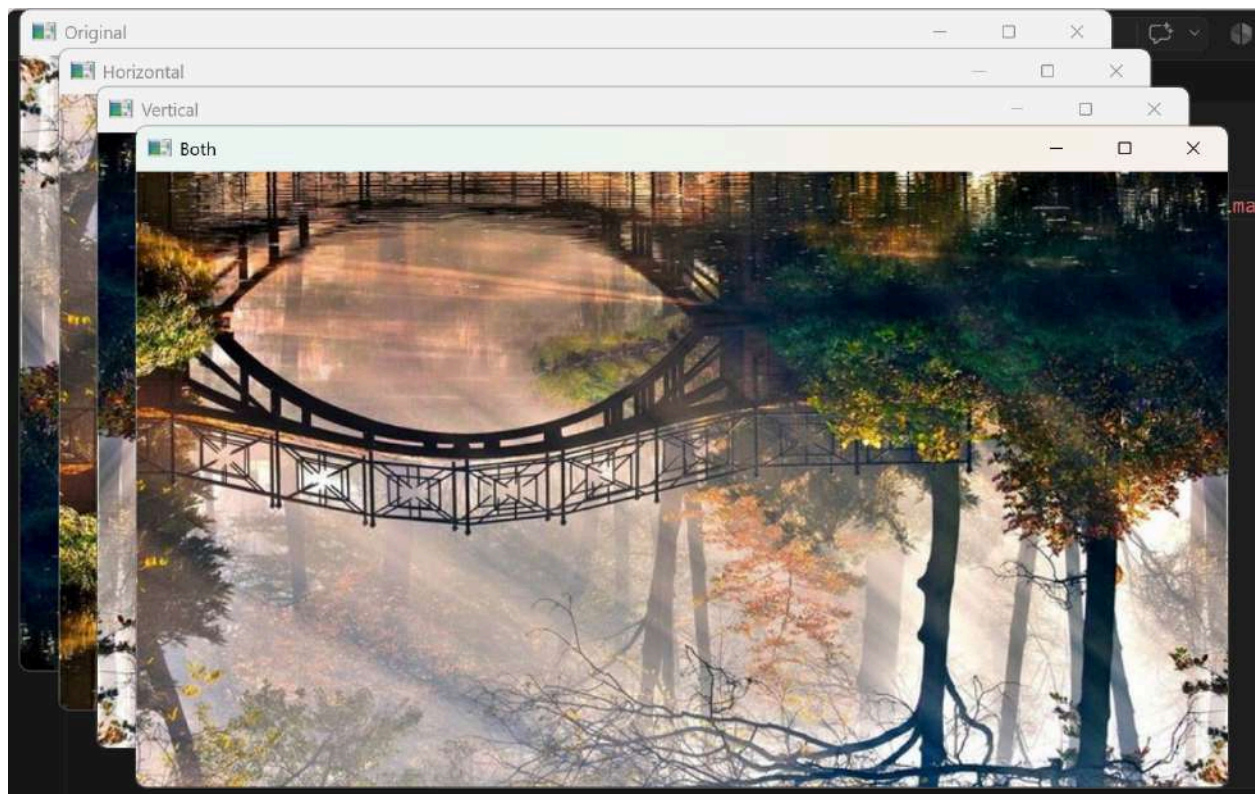




```
File Edit Selection ... OpenCV_Practicals
EXPLORER
OPENCV_PRACTICALS
1_read_write_image.py
2_resize_image.py
3_flip_image.py
4_shapes_text.py
5_translation.py
6_rotation.py
7_threshold.py
8_blur.py
9_morphology.py
10_canny.py
11_video_read_write.py
12_webcam.py
image1.jpg
output.avi
README.md
sample_video.mp4

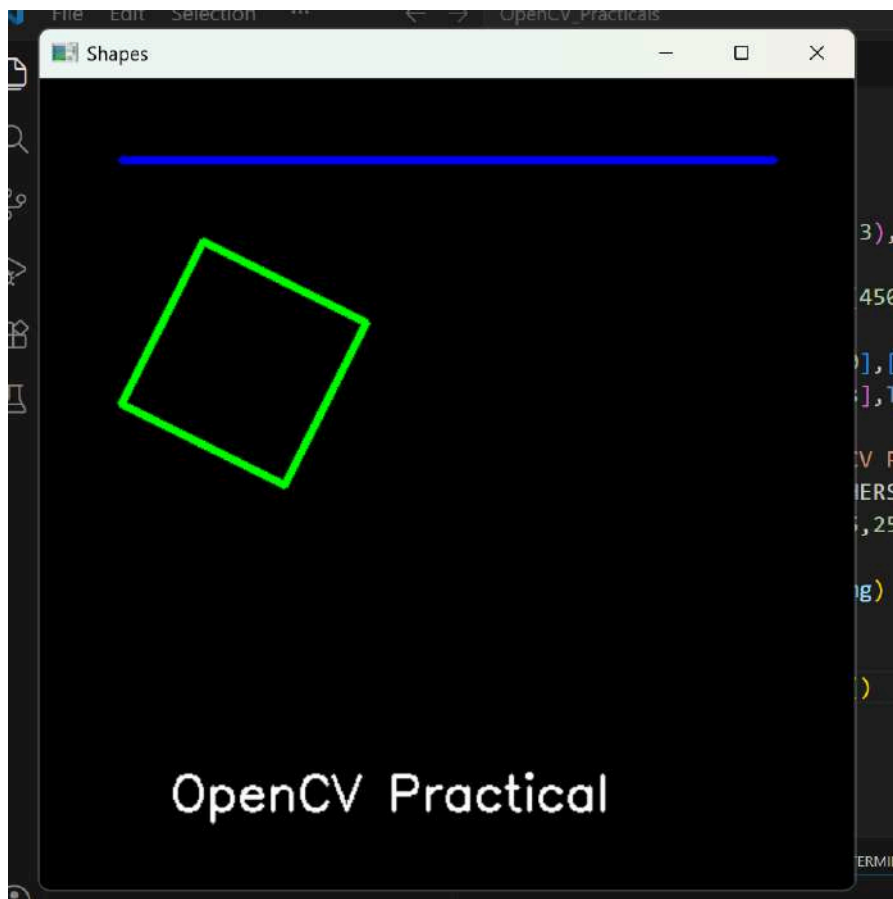
3_flip_image.py 1 X
3_flip_image.py > ...
1 import cv2
2
3 img = cv2.imread(r"C:\Tutedude-Python\OpenCV_Practicals\image1.jpg")
4
5 horizontal = cv2.flip(img,1)
6 vertical = cv2.flip(img,0)
7 both = cv2.flip(img,-1)
8
9 cv2.imshow("Original",img)
10 cv2.imshow("Horizontal",horizontal)
11 cv2.imshow("Vertical",vertical)
12 cv2.imshow("Both",both)
13
14 cv2.waitKey(0)
15 cv2.destroyAllWindows()
```

Output:



```
File Edit Selection ... OpenCV_Practicals
4_shapes_text.py 1 x
4_shapes_text.py > ...
1 import cv2
2 import numpy as np
3
4 img=np.zeros((500,500,3),dtype=np.uint8)
5
6 cv2.line(img,(50,50),(450,50),(255,0,0),3)
7
8 pts=np.array([[100,100],[200,150],[150,250],[50,200]],np.int32)
9 cv2.polylines(img,[pts],True,(0,255,0),3)
10
11 cv2.putText(img,"OpenCV Practical",(80,450),
12             cv2.FONT_HERSHEY_SIMPLEX,
13             1,(255,255,255),2)
14
15 cv2.imshow("Shapes",img)
16
17 cv2.waitKey(0)
18 cv2.destroyAllWindows()
```

Output:

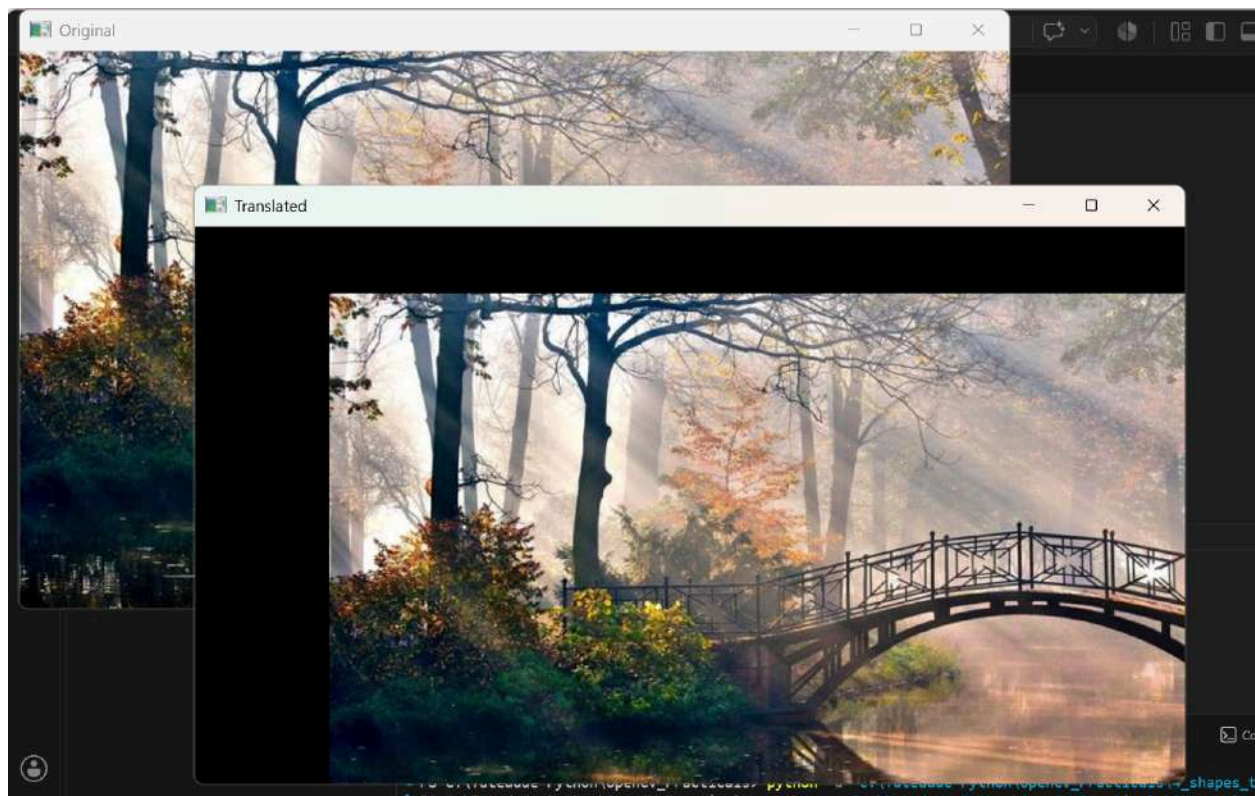


```
File Edit Selection ... OpenCV_Practicals

EXPLORER
└─ OPENCV_PRACTICALS
   ├── 1_read_write_image.py
   ├── 2_resize_image.py
   ├── 3_flip_image.py
   ├── 4_shapes_text.py
   └─ 5_translation.py 1
      ├── 6_rotation.py
      ├── 7_threshold.py
      ├── 8_blur.py
      ├── 9_morphology.py
      ├── 10_canny.py
      ├── 11_video_read_write.py
      ├── 12_webcam.py
      ├── image1.jpg
      ├── output.avi
      ├── README.md
      └── sample_video.mp4

5_translation.py > ...
1  import cv2
2  import numpy as np
3
4  img=cv2.imread("image1.jpg")
5
6  rows,cols=img.shape[:2]
7
8  M=np.float32([[1,0,100],[0,1,50]])
9
10 shift=cv2.warpAffine(img,M,(cols,rows))
11
12 cv2.imshow("Original",img)
13 cv2.imshow("Translated",shift)
14
15 cv2.waitKey(0)
16 cv2.destroyAllWindows()
```

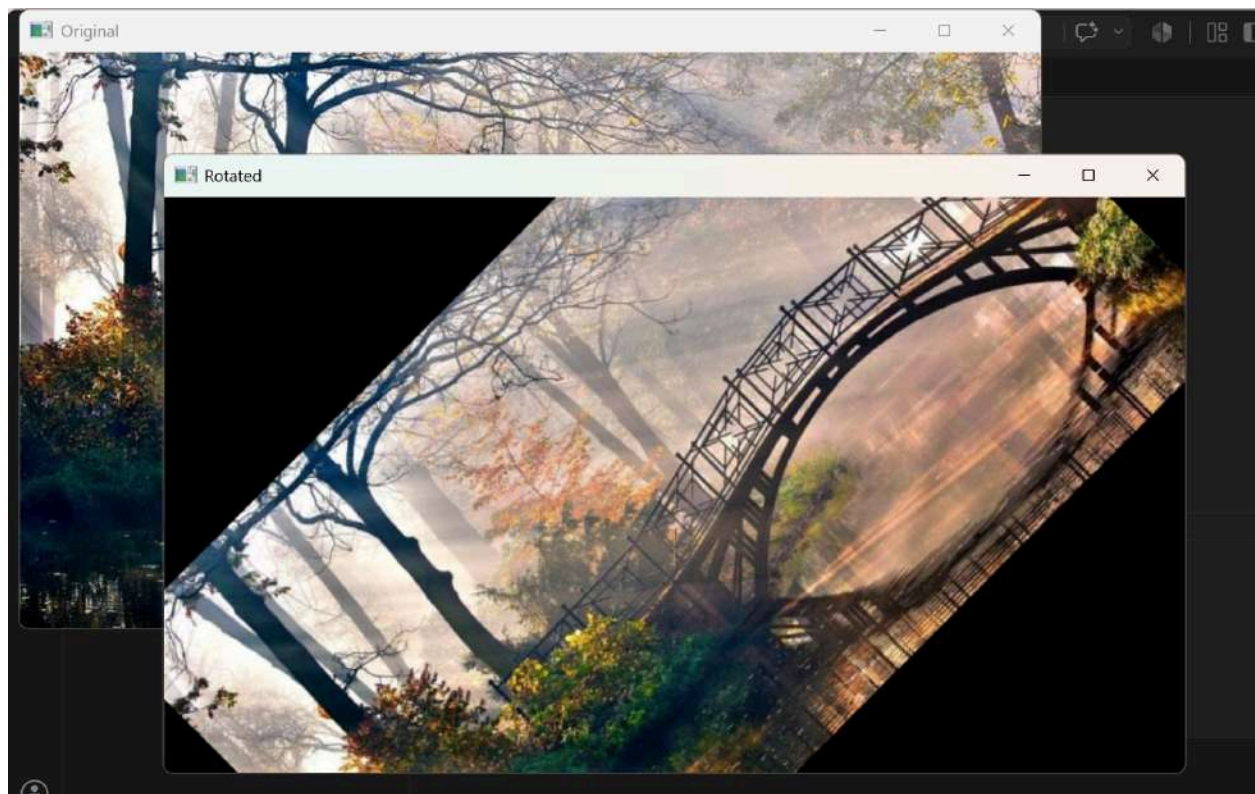
Output:





```
File Edit Selection ... OpenCV_Practicals
6_rotation.py 1 x
6_rotation.py > ...
1 import cv2
2
3 img=cv2.imread("image1.jpg")
4
5 rows,cols=img.shape[:2]
6
7 M=cv2.getRotationMatrix2D((cols/2,rows/2),45,1)
8
9 rotate=cv2.warpAffine(img,M,(cols,rows))
10
11 cv2.imshow("Original",img)
12 cv2.imshow("Rotated",rotate)
13
14 cv2.waitKey(0)
15 cv2.destroyAllWindows()
```

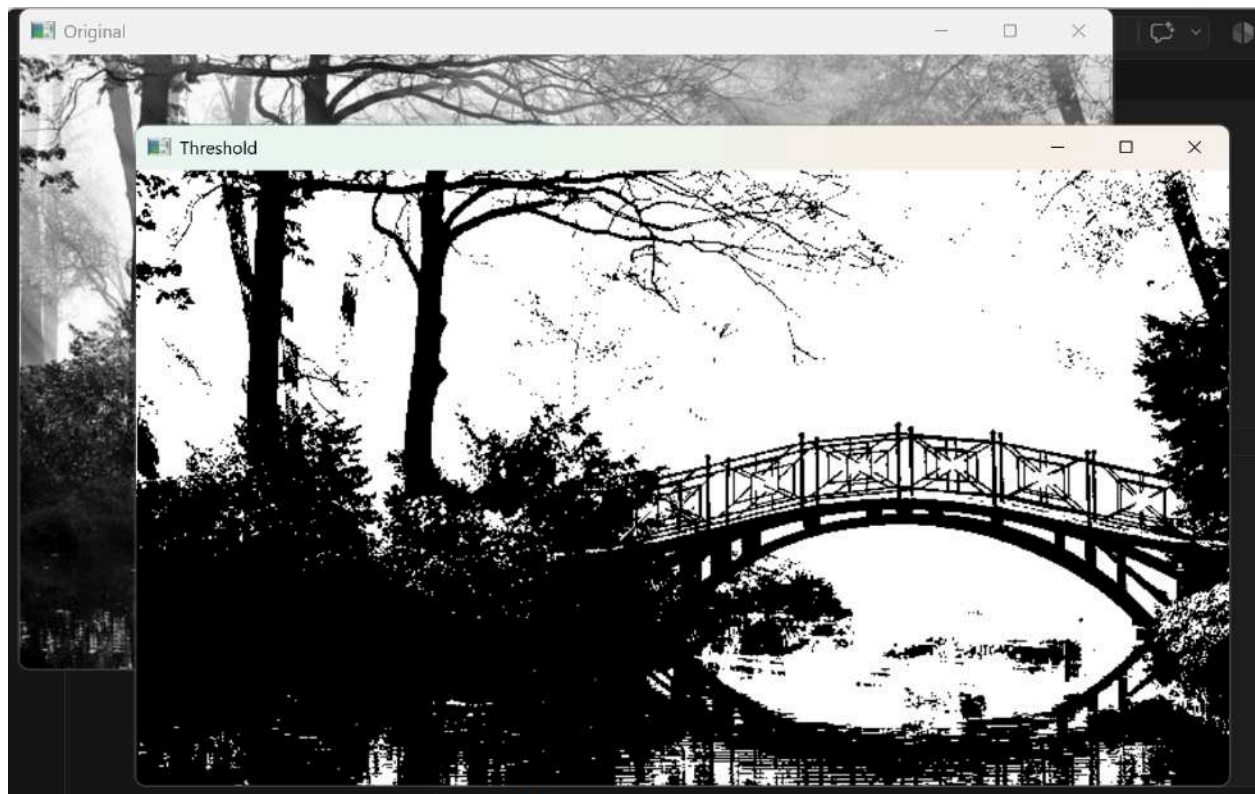
Output:

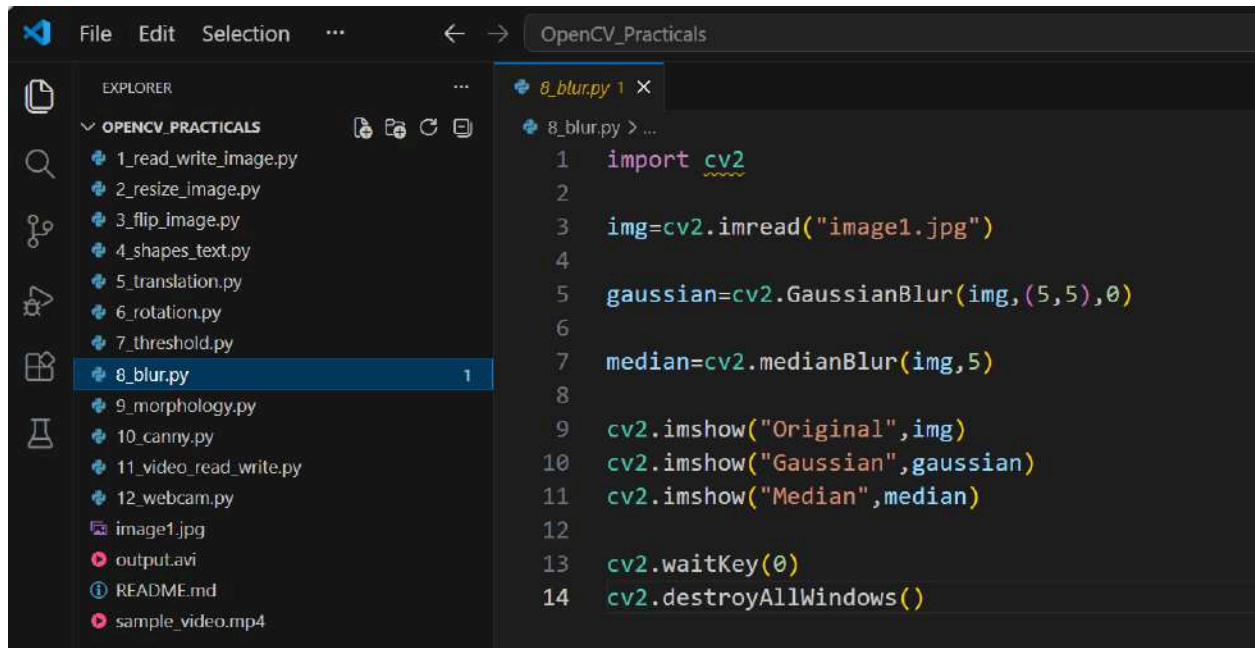


```
File Edit Selection ... OpenCV_Practicals
EXPLORER
OPENCV_PRACTICALS
1_read_write_image.py
2_resize_image.py
3_flip_image.py
4_shapes_text.py
5_translation.py
6_rotation.py
7_threshold.py 1
8_blur.py
9_morphology.py
10_canny.py
11_video_read_write.py
12_webcam.py
image1.jpg
output.avi
README.md
sample_video.mp4

7_threshold.py 1 X
7_threshold.py > ...
1 import cv2
2
3 img=cv2.imread("image1.jpg",0)
4
5 ret,th=cv2.threshold(img,127,255,cv2.THRESH_BINARY)
6
7 cv2.imshow("Original",img)
8 cv2.imshow("Threshold",th)
9
10 cv2.waitKey(0)
11 cv2.destroyAllWindows()
```

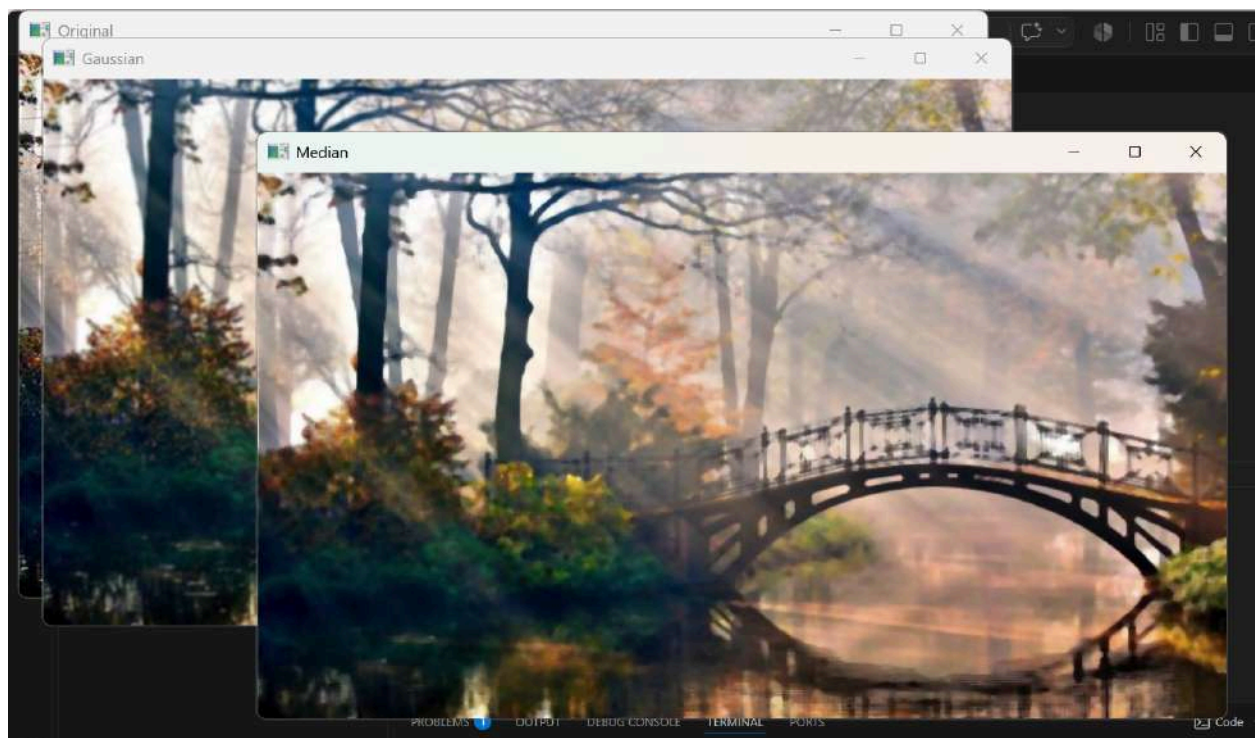
Output:



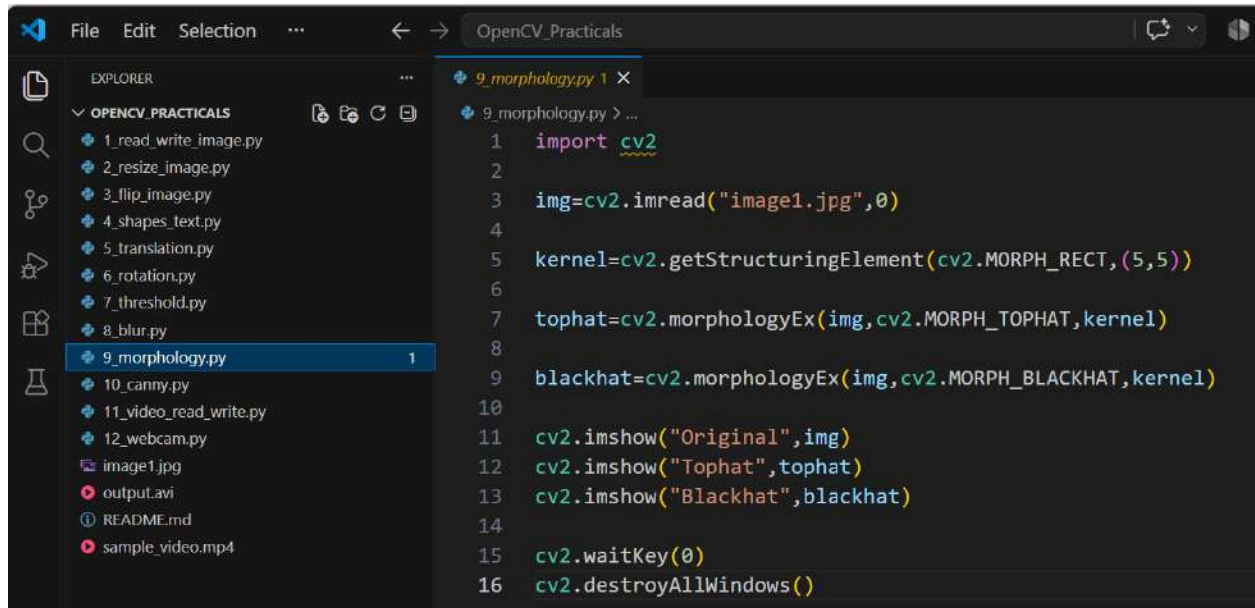


```
1 import cv2
2
3 img=cv2.imread("image1.jpg")
4
5 gaussian=cv2.GaussianBlur(img,(5,5),0)
6
7 median=cv2.medianBlur(img,5)
8
9 cv2.imshow("Original",img)
10 cv2.imshow("Gaussian",gaussian)
11 cv2.imshow("Median",median)
12
13 cv2.waitKey(0)
14 cv2.destroyAllWindows()
```

Output:



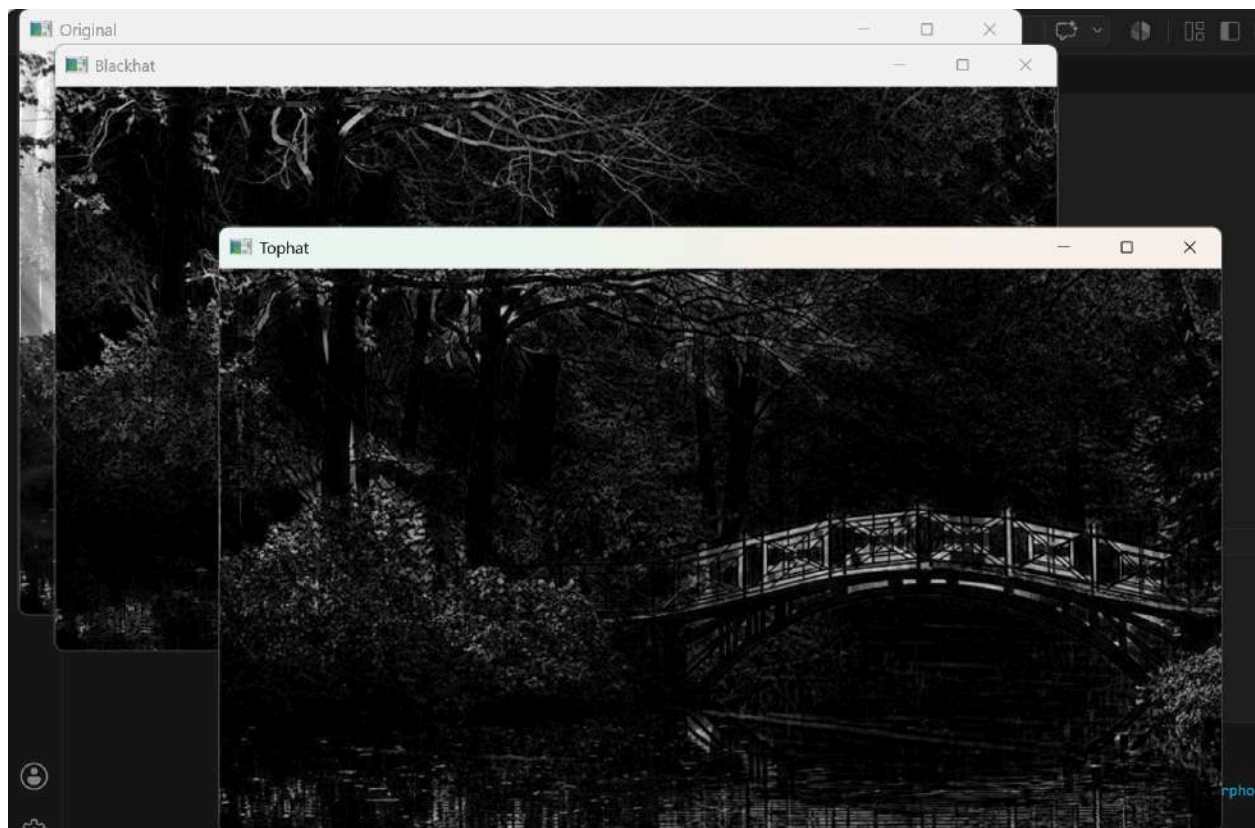




The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a directory named 'OPENCV\_PRACTICALS' containing several Python files. The code editor shows the file '9\_morphology.py' with the following code:

```
1 import cv2
2
3 img=cv2.imread("image1.jpg",0)
4
5 kernel=cv2.getStructuringElement(cv2.MORPH_RECT,(5,5))
6
7 tophat=cv2.morphologyEx(img,cv2.MORPH_TOPHAT,kernel)
8
9 blackhat=cv2.morphologyEx(img,cv2.MORPH_BLACKHAT,kernel)
10
11 cv2.imshow("Original",img)
12 cv2.imshow("Tophat",tophat)
13 cv2.imshow("Blackhat",blackhat)
14
15 cv2.waitKey(0)
16 cv2.destroyAllWindows()
```

Output:

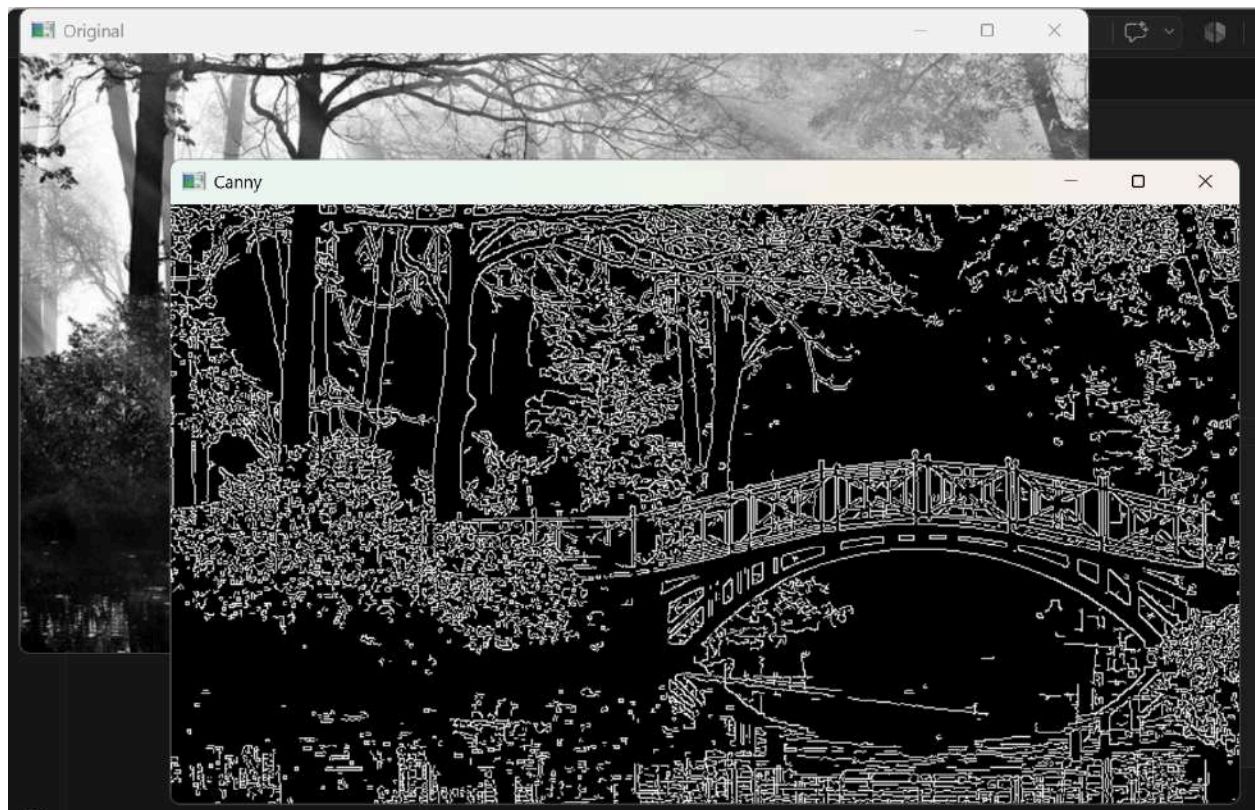


```
File Edit Selection ... OpenCV_Practicals

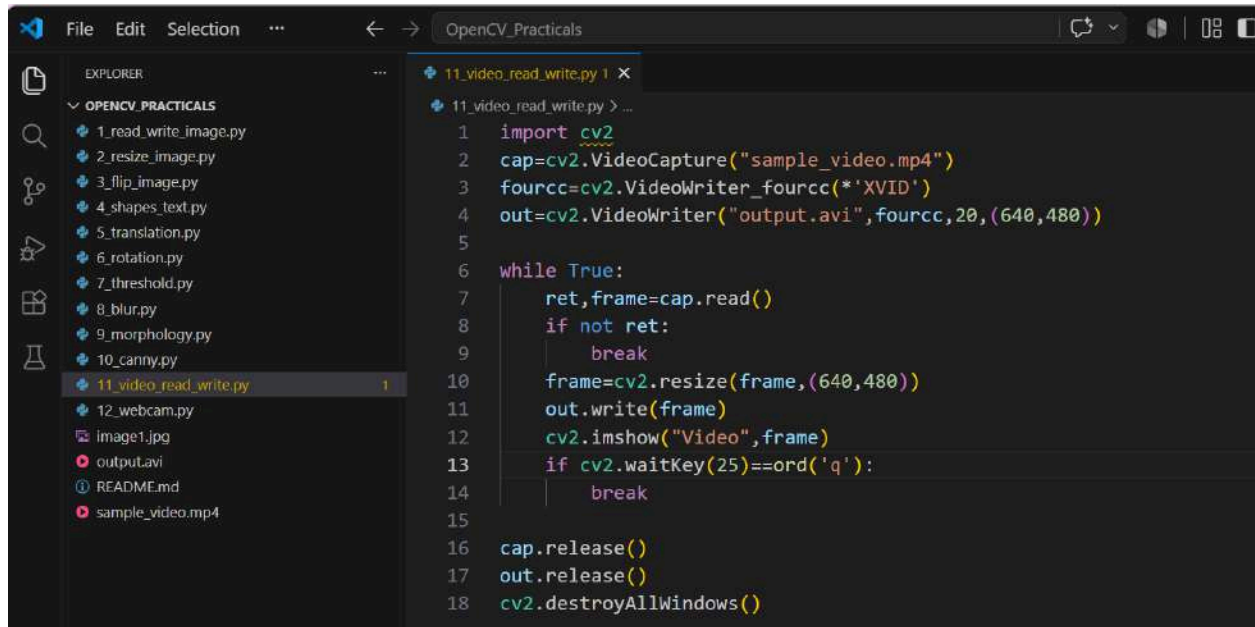
EXPLORER
└─ OPENCV_PRACTICALS
   ├── 1_read_write_image.py
   ├── 2_resize_image.py
   ├── 3_flip_image.py
   ├── 4_shapes_text.py
   ├── 5_translation.py
   ├── 6_rotation.py
   ├── 7_threshold.py
   ├── 8_blur.py
   ├── 9_morphology.py
   └── 10_canny.py 1
       ├── 11_video_read_write.py
       ├── 12_webcam.py
       ├── image1.jpg
       ├── output.avi
       ├── README.md
       └── sample_video.mp4

10_canny.py 1 x
10_canny.py > ...
1  import cv2
2
3  img=cv2.imread("image1.jpg",0)
4
5  edge=cv2.Canny(img,100,200)
6
7  cv2.imshow("Original",img)
8  cv2.imshow("Canny",edge)
9
10 cv2.waitKey(0)
11 cv2.destroyAllWindows()
```

Output:



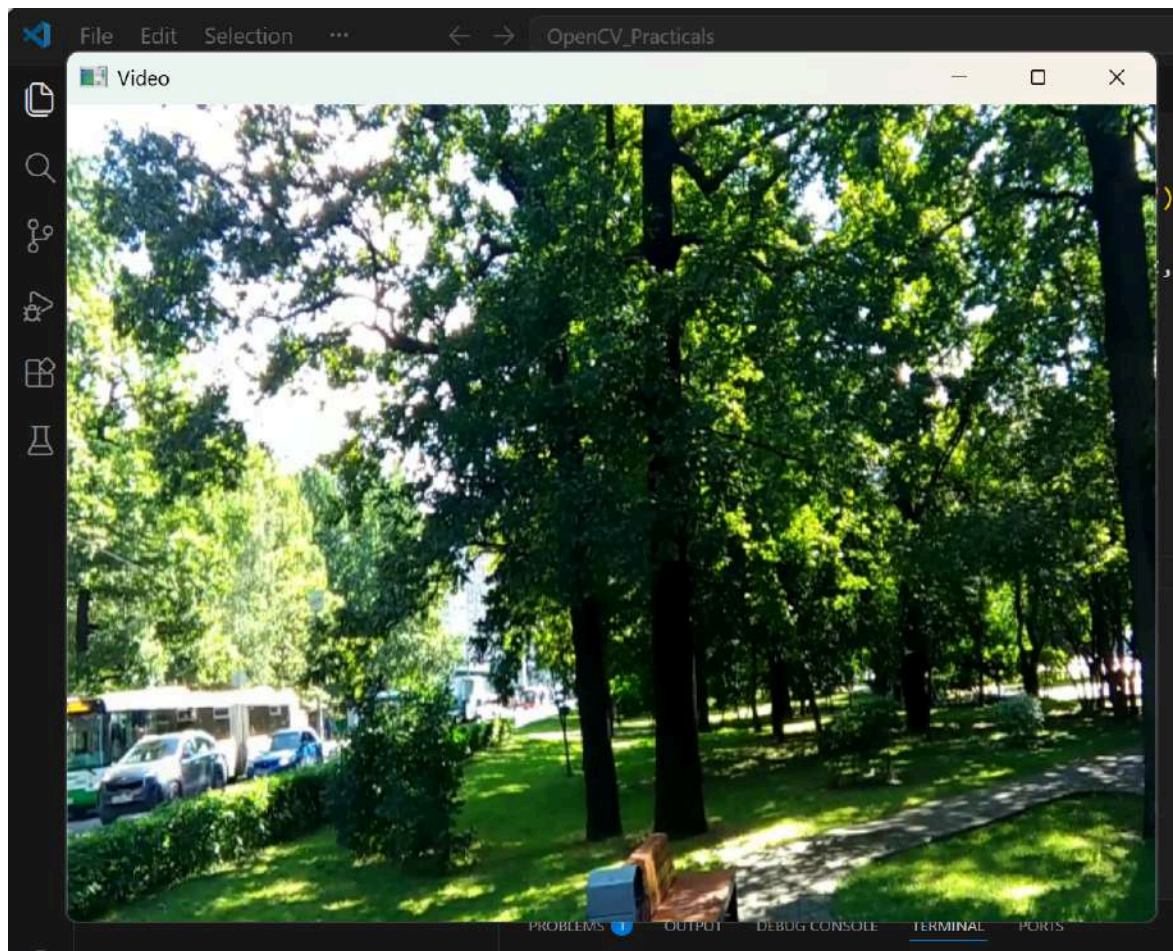




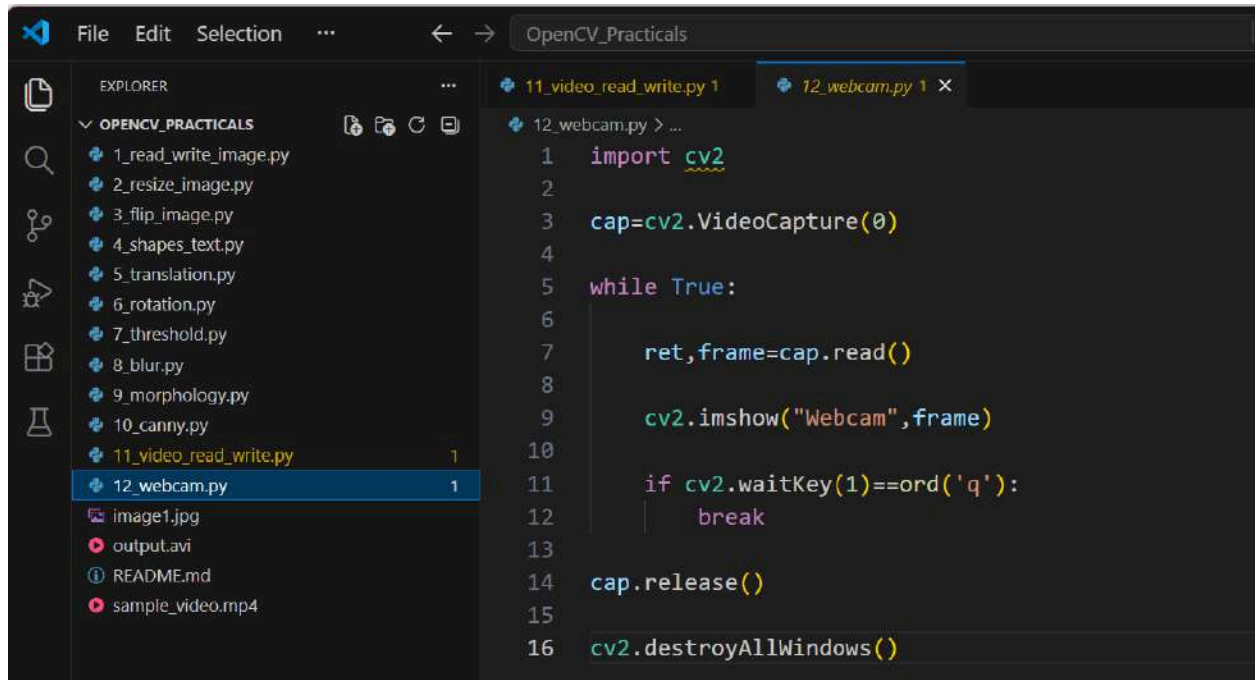
The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer lists several Python files under the name 'OPENCV\_PRACTICALS', including '1\_read\_write\_image.py', '2\_resize\_image.py', '3\_flip\_image.py', '4\_shapes\_text.py', '5\_translation.py', '6\_rotation.py', '7\_threshold.py', '8\_blur.py', '9\_morphology.py', '10\_canny.py', '11\_video\_read\_write.py', and '12\_webcam.py'. The code editor displays the contents of '11\_video\_read\_write.py', which is a Python script for reading and writing video. The script imports 'cv2', captures video from 'sample\_video.mp4', writes it to 'output.avi' with a codec of 'XVID', and displays the video in a window titled 'Video'. The script includes a while loop to read frames, resize them to 640x480, write them, and show them. It also includes a key press check to break the loop and release the video capture and write objects, and finally destroys all windows.

```
1 import cv2
2 cap=cv2.VideoCapture("sample_video.mp4")
3 fourcc=cv2.VideoWriter_fourcc(*'XVID')
4 out=cv2.VideoWriter("output.avi",fourcc,20,(640,480))
5
6 while True:
7     ret,frame=cap.read()
8     if not ret:
9         break
10    frame=cv2.resize(frame,(640,480))
11    out.write(frame)
12    cv2.imshow("Video",frame)
13    if cv2.waitKey(25)==ord('q'):
14        break
15
16 cap.release()
17 out.release()
18 cv2.destroyAllWindows()
```

Output:







```
File Edit Selection ... < -> OpenCV_PRACTICALS

EXPLORER
└─ OPENCV_PRACTICALS
   ├── 1_read_write_image.py
   ├── 2_resize_image.py
   ├── 3_flip_image.py
   ├── 4_shapes_text.py
   ├── 5_translation.py
   ├── 6_rotation.py
   ├── 7_threshold.py
   ├── 8_blur.py
   ├── 9_morphology.py
   ├── 10_canny.py
   ├── 11_video_read_write.py 1
   └── 12_webcam.py 1
       ├── image1.jpg
       ├── output.avi
       ├── README.md
       └── sample_video.mp4

12_webcam.py > ...
1  import cv2
2
3  cap=cv2.VideoCapture(0)
4
5  while True:
6
7      ret,frame=cap.read()
8
9      cv2.imshow("Webcam",frame)
10
11     if cv2.waitKey(1)==ord('q'):
12         break
13
14 cap.release()
15
16 cv2.destroyAllWindows()
```

Output:

